APSP Is Not SETH-Hard Under NSETH

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The All-Pairs Shortest Path problem (APSP) finds the shortest paths between every pair of vertices in a weighted graph. Since this is a function problem, we determine its decision version via *min-plus product*. Given two $n \times n$ matrices A and B, the min-plus product C is defined as

$$C[i,j] = \min_k \{A[i,k] + B[k,j]\} \ .$$

The decision problem is then: given matrices A, B, and C, is C the min-plus product of A and B? This is essentially APSP, which amounts to taking the $(n-1)$-th min-plus power of the adjacency matrix.

We now explain the algorithm of Carmosino et al. [CGI+16], which places APSP in $(N \cap coN)\text{TIME}[\tilde{O}(n^{\frac{3+\omega}{2}})]$. Under NSETH (the Nondeterministic Strong Exponential Time Hypothesis), this implies APSP is not SETH-hard. A note by Bennett [Ben26] explains the analogous algorithm for 3SUM.

Definition 1 (ZWT problem)**.** Given a tripartite directed graph $G(V_1, V_2, V_3, E)$ with $|V_1| = |V_2| = |V_3| = n$ and edge weights in $[-n^a, n^a]$, the *Zero Weight Triangle problem* determines if there exists a triangle whose edge weights sum to zero.

We first show that if the range is small enough, then we can count the number of zero weight triangles efficiently.

Lemma 2. *For a prime p, there is a deterministic algorithm for counting the number of zero weight triangles mod p in time $\tilde{O}(n^\omega p)$.*

Proof. Let A be the weighted adjacency matrix of G. Define matrix B where

$$B[i,j] = \begin{cases} x^{A[i,j] \bmod p} & \text{if } (i,j) \in E \ , \\ 0 & \text{otherwise} \ . \end{cases}$$

Thus each entry of B is a polynomial whose exponent corresponds to the edge weights of A modulo p.

Now compute B^3. By the path-counting property of matrix multiplication, the coefficient of x^j in $B^3[i,i]$ counts the number of triangles through vertex i whose edge weights sum to j. Since every exponent in B lies in $[0, p-1]$, every exponent in B^3 lies in $[0, 3(p-1)]$. The only multiples of p in this range are $0, p$, and $2p$, so summing the coefficients of x^0, x^p, and x^{2p} across all diagonal entries of B^3 counts all triangles with weight $\equiv 0 \pmod p$. Since each triangle is counted once per vertex, dividing this sum by 3 gives us t, the total number of zero weight triangles mod p in G.

The time to compute B^3 is $\tilde{O}(n^\omega p)$, using FFT for multiplying the polynomials and suppressing logarithmic factors. □

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In particular, we use Lemma 2 to verify our nondeterministic guess of the number of false positives is correct.

Lemma 3. *The Zero Weight Triangle Problem is in* $(N \cap coN)\text{TIME}[\tilde{O}(n^{\frac{\omega+3}{2}})]$.

Proof. The NTIME direction is straightforward, guess three vertices and verify their edge weights sum to zero in $O(1)$ time.

To show ZWT is in coNTIME, we verify that there is no zero weight triangle in subcubic time. We prove there exists a prime p^* such that the number of *false positives* (triangles with weight $w \neq 0$ but $w \equiv 0 \pmod{p}$), is smaller than n^3.

Let $\mu = \frac{3-\omega}{2}$ and consider the range $R = [n^\mu, cn^\mu \log n]$ where c is a constant large enough so that R contains at least n^μ primes, which is guaranteed by the Prime Number Theorem.

Assume the instance contains no zero weight triangle, so every triangle has weight $w \neq 0$ with $|w| \leq 3n^a$. Let $p_1, p_2, \ldots, p_k$ be the distinct prime divisors of w that lie in R. Since every prime in R is at least n^μ and their product can't exceed $|w|$, we get

$$(n^\mu)^k \leq p_1 \cdot p_2 \cdots p_k \leq |w| \leq 3n^a \ .$$

Taking logs and solving for k yields,

$$k \leq \frac{\log(3n^a)}{\log(n^\mu)} < \frac{2}{\mu} a \ .$$

Each triangle contributes at most $\frac{2}{\mu}a$ false positives, which is constant. Therefore, the total number of false positives over all primes in R is at most $\frac{2}{\mu}a \cdot n^3$.

Since R contains at least n^μ primes, averaging implies that there exists a prime $p^* \in R$ such that the number of false positives t satisfies

$$t \leq \frac{\frac{2}{\mu}a \cdot n^3}{n^\mu} = \frac{2}{\mu} a \cdot n^{3-\mu} = O(n^{3-\mu}) = O(n^{\frac{\omega+3}{2}}) \ .$$

We now analyze the runtime. Given p^*, Lemma 2 allows us to deterministically compute the number t of triangles with weight $\equiv 0 \pmod{p^*}$ in time

$$\tilde{O}(n^\omega \cdot p^*) = \tilde{O}(n^\omega \cdot n^\mu \log n) = \tilde{O}(n^{\omega+\mu}) = \tilde{O}(n^{\frac{\omega+3}{2}}) \ .$$

We then nondeterministically pick the t false positive triangles and verify each has weight $w \neq 0$ with $w \equiv 0 \pmod{p^*}$, which takes $O(t) = O(n^{\frac{\omega+3}{2}})$ time. Therefore the total runtime is $\tilde{O}(n^{\frac{\omega+3}{2}})$. □

Corollary 4. *APSP* $\in (N \cap coN)\text{TIME}[\tilde{O}(n^{\frac{\omega+3}{2}})]$

Proof. There is a known chain of deterministic fine-grained reductions

$$\text{APSP} \to \text{min-plus product} \to \text{negative weight triangle} \to \text{ZWT} \ .$$

where the reduction from APSP to min-plus product is folklore, and the reductions from min-plus product to negative weight triangle and from negative weight triangle to ZWT are from [VW09] and [WW10] respectively.

The nondeterministic algorithm guesses the min-plus product matrix C and a certificate matrix K where $K[i,j] = \text{argmin}_k\{A[i,k] + B[k,j]\}$, and verifies $C[i,j] = A[i,K[i,j]] + B[K[i,j],j]$ for all i, j in $O(n^2)$ time, which proves no entry of C is too large.

To verify no entry of C is too small, construct a complete tripartite graph with weight matrices $-A$ between V_1 and V_2, $-B$ between V_2 and V_3, and C between V_1 and V_3. A triangle (i, k, j) has total weight $C[i, j] - A[i, k] - B[k, j]$, which is negative if and only if some entry of C is too small. Certifying no such triangle exists then follows from the co-nondeterministic ZWT algorithm of Lemma 3 in time $\tilde{O}(n^{\frac{\omega+3}{2}})$.

Modifying these reductions slightly to be nondeterministic gives the optimal bound of $\tilde{O}(n^{\frac{\omega+3}{2}})$. □

References

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